Hide menu

Orthogonality
Overdetermined System: Best approximation
Best Approximation equals Orthogonality leads to Normal Equations
Example: Approximate Curve Fitting
Assignment 4
Quiz: Linear Least Squares
Submitted

Linear Least Squares

Submit your assignment

Due Jan 21, 11:59 PM IST

Receive grade

To Pass 80% or higher

Like Dislike Report an issue

Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

Try again

Your grade 100%

View Feedback

We keep your highest score

1. **1 / 1 point**

What is the length $\|\mathbf{v}\|$ of the given $\mathbf{v}$?

$$\left\| \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\| =$$

Correct

2. Inner product between two vectors **1 / 1 point**

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = ?$$

Correct

3. You enter a room at the southwest corner (origin x=0, y=0). A light bulb is located in the middle of the room at x=3 feet east and y=4 feet north of the entrance, 7 feet above the floor. You must change the bulb, and need a ladder to reach the bulb. Where do you place the ladder on the floor to be closest to the bulb so you can reach the bulb? **1 / 1 point**

Correct

4. Q: **1 / 1 point**

In the SVD, the U & V matrices are called "orthogonal" because their columns are all mutually orthogonal to each other and have length 1. These are exactly the matrices that preserve the length of any vector to which they are multiplied: $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for any vector $\mathbf{x}$ (where $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^T\mathbf{x}} = \sqrt{x_1^2 + \cdots x_n^2}$ = the length of $\mathbf{x}$). Which of the following matrices have this property of being orthogonal?

Correct